

考研精读 | Economist



Dinosaur gastroliths

恐龙胃石

难度评级: ★★★★★

Rock-solid evidence for the origins of birds

鸟类起源的坚实证据

【 Para.1 】

How do you know what a dinosaur ate? Guts **rot** away after death, so the details of dinosaur digestion remain **obscure**.

The best indicators are teeth. Sharp, **pointed** teeth probably **ripped chunks** of flesh off **prey** for swallowing whole.

Robust grinding surfaces suggest lots of chewing was involved. Yet many dinosaurs belonging to a group called the **theropods**, which **gave rise to** birds, lost their teeth altogether and **evolved beaks**. This has made it hard to work out what they ate and how they digested it.

如何知晓恐龙以何为食？内脏在死亡后已腐烂，因而恐龙消化过程的细节始终扑朔迷离。最可靠的线索当属牙齿。尖利锋锐的牙齿，很可能用来从猎物身上撕下肉块，然后整块吞咽；而结实的研磨面，则暗示着大量咀嚼行为的存在。然而，恐龙中的一支——兽脚

August 15TH- 21ST | Science & technology

文章导读：

本研究通过对比现代主龙类生物与恐龙的胃石，探究恐龙进食与演化规律。食草鸟臀目、长颈蜥脚类恐龙多棱角胃石，依靠颞部咀嚼或胃部发酵消化。无齿具喙的兽脚类恐龙拥有圆润胃石，食物研磨功能由颌部转向胃部。这一体态重心变化，助力鸟类最终演化出飞行能力。

gastrolith n. /'gæstrəli θ/ a stone in the stomach; gastric calculus 胃石

rot v. /rɔ:t/ to decay, or make sth decay, naturally and gradually (使) 腐烂，腐败变质 【考研大纲词汇】

rotting leaves 渐渐腐烂的叶子

The window frame had **rotted** away completely. 窗框已经完全烂掉了。

obscure adj. /əb'skjʊə(r)/ difficult to understand 费解的；难以理解的 【考研大纲词汇】

pointed adj. having a sharp end 尖的；有尖头的 【考研大纲词汇】

a **pointed chin** 尖下巴

pointed teeth 尖尖的牙齿

rip v. to tear sth or to become torn, often suddenly or violently (突然或猛烈地) 撕破，裂开 【考研大纲词汇】

I **ripped** my jeans on the fence. 我的牛仔褲在柵欄上划破了。

chunk n. chunks of something are thick, solid pieces of it. 厚块

prey n. /prei/ an animal, a bird, etc. that is hunted, killed and eaten by another 猎物 【考研大纲词汇】

robust adj. /rəʊ'bʌst/ strong; able to survive being used a lot and not likely to break 结实的；耐用的；坚固的 【考研大纲词汇】

grind v. /graɪnd/ ~ sth (down/up), ~ sth (to/into sth) to break or crush sth into very small pieces between two hard surfaces or using a special machine 磨碎；碾碎；把...磨成粉 【考研大纲词汇】

to **grind** coffee/corn 将咖啡/谷粒磨成粉

theropod n. /' θ iərəpɒd/ any bipedal carnivorous saurischian dinosaur of the suborder Theropoda, having strong hind legs and grasping hands. They lived in Triassic to Cretaceous times and included tyrannosaurs and megalosaurs 兽脚亚目恐龙

give rise to phrase to cause sth to happen or exist 使发生（或存在） 【考研大纲词汇】

The novel's success **gave rise to** a number of sequels. 这部小说的成功带来了一系列的续篇。

evolve v. /i'vɒlv/ ~ (from sth) (of plants, animals, etc. 动植物等) to develop over time, often many generations, into forms that are better adapted to survive changes in their environment 进化；进化形成 【考研大纲词汇】

亚目，最终演化出了鸟类——却完全丧失了牙齿，转而演化出喙部。这一变化使得探究其食性与消化方式变得困难重重。

【 Para.2 】

There is, however, a second clue. Lots of dinosaurs had stones, known as gastroliths, in their stomachs. These are assumed to have assisted with digestion, and thus have their own **tales** to tell. And modern birds and **crocodilians** (which are both, like dinosaurs, members of a larger **evolutionary** group called the **archosaurs**) often have gastroliths, too.

不过，尚有第二条线索。许多恐龙的胃中存有石块，即所谓“胃石”。这些石块普遍被认为有助于消化，因而它们本身亦蕴藏着可供解读的信息。而现代鸟类与鳄类（二者与恐龙同属一个更大的演化支系，即祖龙类）体内亦常见胃石。

【 Para.3 】

Takasaki Ryuji at Okayama University of Science, in Japan, decided to investigate. In a study published in Paleobiology, he and his colleagues examined the gastroliths of 46 modern archosaur species (42 birds and four crocodilians) and compared them with those from 16 species of dinosaur.

日本冈山理科大学的高崎龙司（Takasaki Ryuji）决定就此展开研究。在发表于《古生物学》期刊的一项研究中，他与同事检视了46种现代祖龙类（包括42种鸟类和4种鳄类）的胃石，并将其与16种恐龙的胃石加

beak n. /bi:k/ the hard pointed or curved outer part of a bird's mouth 鸟喙

The gull held the fish in its **beak**. 海鸥嘴里叼着鱼。

tale n. an exciting spoken description of an event, which may not be completely true (精彩但不一定完全真实的) 讲述，叙述 【考研大纲词汇】

crocodilian n. /ˌkrɒkəˈdɪliən/ any large predatory reptile of the order Crocodilia, which includes the crocodiles, alligators, and caymans. They live in or near water and have a long broad snout, powerful jaws, a four-chambered heart, and socketed teeth 鳄目动物

evolutionary adj. /ˌevəˈluːʃənəri/ connected with evolution; connected with gradual development and change 进化的；演变的；逐渐发展的

archosaur n. /ˈɑːrkəˌsɔːr/ a reptile of a large group that includes the dinosaurs and pterosaurs, represented today only by the crocodilians [动,古生物] 古蜥, 祖龙

herbivore n. /ˈhɜːbrɪvɔː(r)/ any animal that eats only plants 食草动物；草食动物

omnivore n. /ˈɑːmnɪvɔːr/ an animal or a person that eats all types of food, especially both plants and meat 杂食动物；杂食的人

carnivore n. /ˈkɑːnɪvɔː(r)/ any animal that eats meat 食肉动物

goose n. a bird like a large duck with a long neck. Geese either live wild or are kept on farms 鹅 【考研大纲词汇】

wear v. to become, or make sth become thinner, smoother or weaker through continuous use or rubbing 磨损；消耗；用旧 【考研大纲词汇】

The carpets are starting to **wear**. 地毯渐渐磨坏了。

The stones have been **worn** smooth by the constant flow of water. 不停的流水把这些石头冲刷得很光滑。

pebble n. /ˈpebl/ a smooth, round stone that is found in or near water 鹅卵石；砾石 【考研大纲词汇】

angular n. /ˈæŋɡjələ(r)/ having angles or sharp corners 有棱角的；有尖角的

ornithischian n. /ˌɔːnɪˈθɪskɪən/ any dinosaur of the order Ornithischia; a bird-hipped dinosaur 鸟臀目恐龙

Triceratop n. /ˈtraɪˈce.rə.tɒps/ 三角龙

(tri 表示“三”，cerat 表示“角”，ops 表示“脸”，合起来就是“有三只角的那张脸”)

retain v. /riˈteɪn/ to keep sth; to continue to have sth 保持；持有；保留；继续拥有 【考研大纲词汇】

SYN preserve

to **retain** your independence 保持独立

jaws n. [pl.] the mouth and teeth of a person or an animal 口部；嘴 【考研大纲词汇】

sauropod n. /ˈsɔːrəpɒd/ any herbivorous quadrupedal saurischian dinosaur of the suborder Sauropoda, of Jurassic and Cretaceous times, including the brontosaurus, diplodocus, and titanosaurs. They had small heads and long necks and tails and were partly amphibious 蜥脚类动物

Diplodocus n. /dɪˈplɒdəkəs/ a very large dinosaur with a long thin neck and tail 梁龙（颈部和尾巴细长的大恐龙）

以对比。

【 Para.4 】

The birds included **herbivores**, **omnivores** and **carnivores**.

The crocodilians were all carnivorous. The shapes of the stones varied. As might be expected, if an animal had a highly muscular stomach (ducks and **geese**), its gastroliths tended to have been **worn** into the sorts of rounded shapes displayed by beach **pebbles**. Weaker stomachs (seabirds and crocodiles) contained more **angular** stones.

研究所涉及的鸟类涵盖草食、杂食和肉食种类；鳄类则全部为肉食性。这些胃石的形态各异。可以想见，若某动物拥有肌力发达的胃（如鸭和鹅），其胃石往往会被磨蚀为河滩卵石般圆润的形态；而胃部肌肉力量较弱的动物（如海鸟和鳄鱼），胃中则多为有棱有角的石块。

【 Para.5 】

Looking at the dinosaurs, Dr Takasaki and his colleagues found that **ornithischians** (think **Triceratops**), a group which their teeth suggest were herbivores, **retained** angular gastroliths. These creatures, this implies, relied mostly on their **jaws** to grind up their food, leaving their stomachs to apply merely the finishing touches.

在检视恐龙时，高崎博士及其团队发现，鸟臀目恐龙（如三角龙）——其牙齿特征表明其为草食性动物——胃中保留了棱角分明的胃石。这意味着，此类生物主要依靠颞部研磨食物，胃部仅作最后加工。

puzzle n. something that is difficult to understand or explain 不解之谜；疑问 【考研大纲词汇】
SYN mystery

strip v. ~ sth (off), ~ A (off/from B), ~ B (of A) to remove a layer from sth, especially so that it is completely exposed 除去，剥去（一层）；（尤指）剥光 【考研大纲词汇】
Deer had stripped the tree of its bark. 鹿啃掉了树皮。

cervically adv. 在颈部方向地/方式地

elongated adj. /'i:lɒŋɡeɪtɪd/ long and thin, often in a way that is not normal 细长的；拉得又细又长的
Modigliani's women have strangely elongated faces. 莫迪里阿尼画中的妇女都长着奇长无比的脸。

ungulate n. /'ʌŋɡjʊlət/ any animal which has hoofs, such as a cow or horse 有蹄类动物

Tarbosaurus n. 特暴龙，属于暴龙科，是晚白垩世生活在亚洲的大型暴龙科恐龙，与霸王龙亲缘关系很近但属于独立物种，化石主要发现于蒙古和中国，是当时亚洲生态系统中的顶级掠食者

hypercarnivore n. 超级食肉动物（进食食物中有 70% 为肉类的动物）

Tyrannosaurus n. /tɪˈrænəˈsɔːrəs/ any large carnivorous bipedal dinosaur of the genus Tyrannosaurus, common in North America in upper Jurassic and Cretaceous times; suborder Theropoda (theropods) 暴龙；晚侏罗纪和白垩纪中在北美常见

be endowed with sth phrase to naturally have a particular feature, quality, etc. 天生赋有，生来具有（某种特性、品质等）
She was endowed with intelligence and wit. 她天资聪颖。

centralise v. /'sentrəlaɪz/ 使集中；使成为...的中心

【 Para.6 】

Long-necked **sauro pods** (think **Diplodocus**) also had angular gastroliths. That is more of a **puzzle**, for these animals had pencil-like teeth using which, it is assumed, they **stripped** the leaves off tree branches that less **cervically elongated** animals were unable to reach. The explanation, Dr Takasaki suspects, is that like those of modern **ungulates**, sauro pods' stomachs depended mainly on fermentation to break food down.

长颈的蜥脚类恐龙（如梁龙）同样拥有棱角分明的胃石。这却更令人费解，因为这些动物的牙齿形如铅笔，据推测，它们用这些牙齿将枝叶从那些脖颈较短的动物无法触及的树枝上捋下。高崎博士推测，其原因在于，蜥脚类恐龙的胃与现代有蹄类动物相似，主要依赖发酵作用来分解食物。

【 Para.7 】

Stones in the guts of theropods tell yet another story. **Tarbosaurus**, a **hypercarnivore** similar to **Tyrannosaurus**, and similarly **endowed with** sharp, pointed teeth, had angular gastroliths. However, theropods with beaks (of which science recognises at least four groups) had rounded ones. Indeed, Dr Takasaki saw a clear relation between tooth loss and stone roundness, as the work of grinding up food was transferred from jaws to stomach.

兽脚类恐龙消化道中的胃石则呈现另一番图景。特暴龙——一种与暴龙相似、同样生有尖利牙齿的超级肉

►今日句子分析

Para 6. That is more of a puzzle, for these animals had pencil-like teeth using which, it is assumed, they stripped the leaves off tree branches that less cervically elongated animals were unable to reach.



句子分析：

主句：

主语-That（指代前一句“长颈蜥脚类也有棱角胃石”这一事实）；

系动词- is；表语-more of a puzzle,

（more of a + n. 结构，意为“在更大程度上是一个谜”）

原因状语从句：

连词-for；主语-these animals；谓语-had；宾语- pencil-like teeth；状语- using ...

定语从句：

引导词，指代teeth- which；

（插入句：主语- it；谓语-is assumed,）

主语-they；谓语-stripped；宾语-the leaves；状语- off tree branches；

定语从句：

引导词，指代tree branches，作 reach 的宾语- that；主语-less cervically elongated animals；系动词- were；表语-unable；补语- to reach.

翻译解析：长的复合句，首先要拆分结构：主干 + 原因状语从句 + 多层定语从句。

其次，解决翻译难点：more of a...是英文特有结构，表示“在更大程度上是.....而非其他”，通常不直译，译为“更令人.....”或“与其说是.....不如说是.....”；cervically elongated 属于专业性词汇，意译为“脖子长的/脖子短的”，化术语为通俗表达更合适。

食者——胃中为棱角分明的胃石。然而，长有喙部的兽脚类恐龙（科学界公认至少有四个类群）则拥有圆润的胃石。事实上，高崎博士观察到了牙齿退化与胃石圆润程度之间的明确关联，研磨食物的工作，正从颌部逐渐转移至胃部。

【 Para.8 】

That shift may have helped make possible the evolutionary success of birds. Teeth are heavy, as are the jaw muscles those teeth require to do their job. This puts a lot of weight near an animal's front. Shifting the job of grinding, and the muscles and hard surfaces involved, to the stomach **centralises** an animal's centre of gravity in a way that would make it easier for it to rise from the ground.

这一转变，或许为鸟类演化的成功铺平了道路。牙齿很沉重，而且驱动牙齿工作所需的颌部肌肉则更添负担。这给动物前部带来了不小的重量。将研磨功能及其所需的肌肉与坚硬结构转移至胃部，能使动物的重心向躯干中心集中，从而更易于从地面跃起升空。

【 Para. 9 】

Curiously, it is clear from the fossil record that the very earliest birds of all did not have beaks. But the fact that so many of their theropod relatives did, and that birds themselves evolved them at least twice, suggests the transition was easy. It may thus be that part of the secret to taking successfully to the skies was, quite literally, having the stomach for it.

➤ 写作积累

Para.2 These are assumed to have assisted with digestion, and thus have their own tales to tell.

be assumed to... “被认为..... / 据推测.....”，通常表示某种观点是“被大家普遍接受”或“基于现有证据推测”的，但并不是绝对的事实。这个表达比 is known to（已知）要严谨，比 is thought to（被认为）稍微正式一点。此外，assume 也常见于另一种句型，即 it is assumed that..., 例句：In America, it is assumed that a student's daily life is as important as his learning experience. 在美国，人们认为一个学生的日常生活和他的学习经验一样重要。

原句这里作者用 are assumed to have assisted 表示“人们普遍认为/推测它们曾经帮助过消化”，相对而言会显得更加客观、严谨。

assist with... “协助..... / 对.....有帮助” assist 是 help 的正式用语，搭配 with 后面通常接名词或动名词，表示“在某方面提供帮助”。

它还有另一种用法，即 assist sb in doing sth “协助某人做某事”，例：Our sales representatives can assist you in selecting suitable investments. 我们的销售代表在选择合适的投资方面能帮到您。

have their own tales to tell 字面意思为“有着它们自己的故事要讲述”，实际引申为“蕴含的经历、历史或秘密”

耐人寻味的是，化石记录清晰地表明，最早的鸟类其实并无喙部。然而，它们的众多兽脚类近亲早已演化出喙，而鸟类自身又至少独立演化出喙两次，这一事实表明，这一转变并非难事。因此，鸟类得以成功飞向天空，其部分奥秘或许恰恰在于它们拥有能够承载这一切的“胃”。